

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	CHERYL CHEONG KAH VOON
IC No. / Matric No.	040322040352/A23CS0060
Year / Program	2/SECPH
Section	Section 01
Lecturer Name	MS. ROZILAWATI BINTI DOLLAH @ MD. ZAIN
Case Study	WBL Project
System Name	Koperasi Kada Online System
Group Name	Technicrab

SECTION A (SYSTEM'S OVERVIEW)		
1.1	Overall Description	
		<< <i>Describe the system's overview.</i> >>
1.2	Project Weaknesses or Improvement	
		<< <i>List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses.</i> >>
1.3	Proposed Module	
		<< <i>Describe a new module to overcome the project's weaknesses.</i> >>
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	Conceptual ERD	
		<< <i>Draw a conceptual ERD to represent the proposed module.</i> >>
2.2	Logical ERD	
		<< <i>Provide logical ERD of your module in the proposed system.</i> >>
2.3	Data Dictionary for Logical ERD	
		<< <i>Provide the data dictionary based on the logical ERD in Subsection 2.2.</i> >>
2.4	Relational Database Schema	
		<< <i>Provide all relations database schema of your proposed module.</i> >>
2.5	Normalization	
		<< <i>Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved.</i> >>

Section A (System's Overview)

1.1 Overall Description

Our client is Koperasi Kada Kelantan Berhad, a Lembaga Kemajuan Pertanian Kemubu (KADA) subsidiary. The system that we proposed is the Koperasi KADA Online System. The Koperasi KADA Online System proposed is a web-based, fully integrated digital platform designed to modernize and streamline the cooperative's operations, replacing the outdated manual system which is time-consuming, prone to errors and heavily reliant on human resources. The system mainly functions as recording, maintaining and managing the data regarding the management of the client's company, such as information about members and finances to ease the process of management of the client to release them from the paperwork management system and facilitate the communication of members with Koperasi KADA. This system has three key modules, which are the employee module, member module and admin module. Each module has functions that are specific to its user group, addressing the cooperative's operating demand and improving efficiency.

The employee module facilitates the membership application process for employees. Employees can apply for membership and check their status of membership application. The member module allows members to manage their communication with the cooperative. Key features include members viewing financial status, applying for a loan, managing their profiles and viewing loan application status. The admin module allows administrative staff to monitor and manage system operations. This includes approving membership registration and loan applications and creating reports for better decision-making and performance tracking.

The initiation of this project stems from several critical issues faced by Koperasi KADA Kelantan Berhad due to its reliance on a manual system:

- Manual processes like member registration and loan application are inefficient and time-consuming causing delays in service.
- Manual data management results in high error rates, including missing documents, incorrect records and overlooked member information.
- Members cannot check their loan applications or financial details in real-time, resulting in them having to contact staff directly or wait for an email from the company to obtain the current status, which is inconvenient and time-consuming.

- Storing sensitive member data in physical formats raises the risk of data breaches, unauthorized access, which all staff in the office can access to see the document, and information loss due to mishandling or disasters.
- The cooperative's current system relies mainly on staff to manually process registration, financial statements and loans, resulting in reduced productivity and scalability.

Group Members' Name	Matric No	Module
Cheryl Cheong Kah Voon	A23CS0060	Member module (financial status & manage profile)
Chau Ying Jia	A23CS0213	Member module (apply for loan & loan application status)
Neo Li Xin	A23CS0253	Admin module (generating reports)
Lau Yee Wen	A23CS0253	Employee Module (apply for membership & membership application status)
Woo Cheng Shuan	A23CS0283	Admin module (membership and loan application approval)

1.2 Project Weaknesses or Improvements

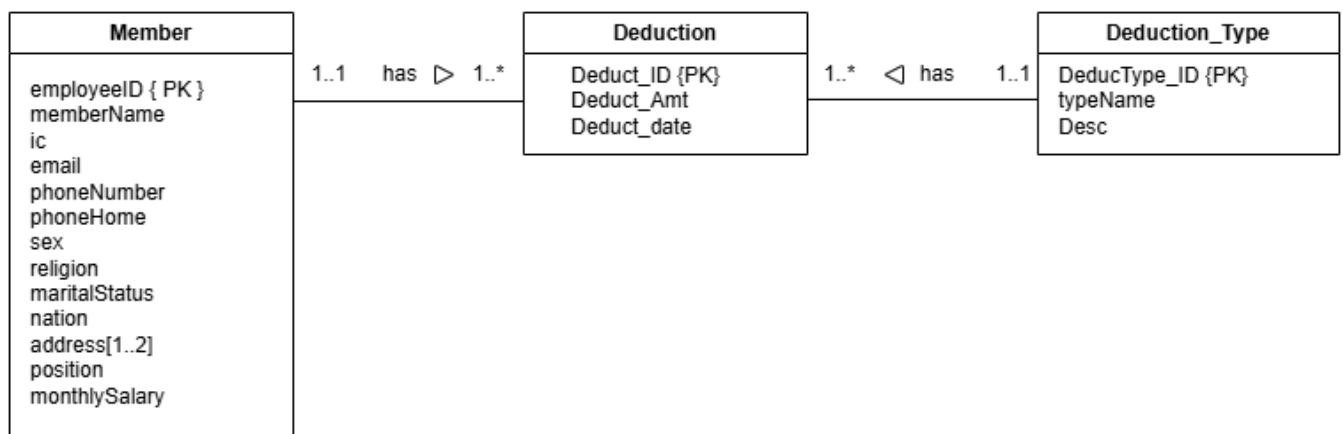
One of the current system's weaknesses is that the admin cannot create or customize new types of salary deductions dynamically. The system is limited to predefined deduction types, reducing its flexibility to adapt to changing payroll requirements. This lack of customizability also causes inefficiency as any change requires direct database editing or human adjustments. This process is both time-consuming and error-prone. To solve this, a customized deduction module is proposed, allowing the admin to generate new deduction types and dynamically link them to members, improving usability and scalability.

1.3 Proposed Module

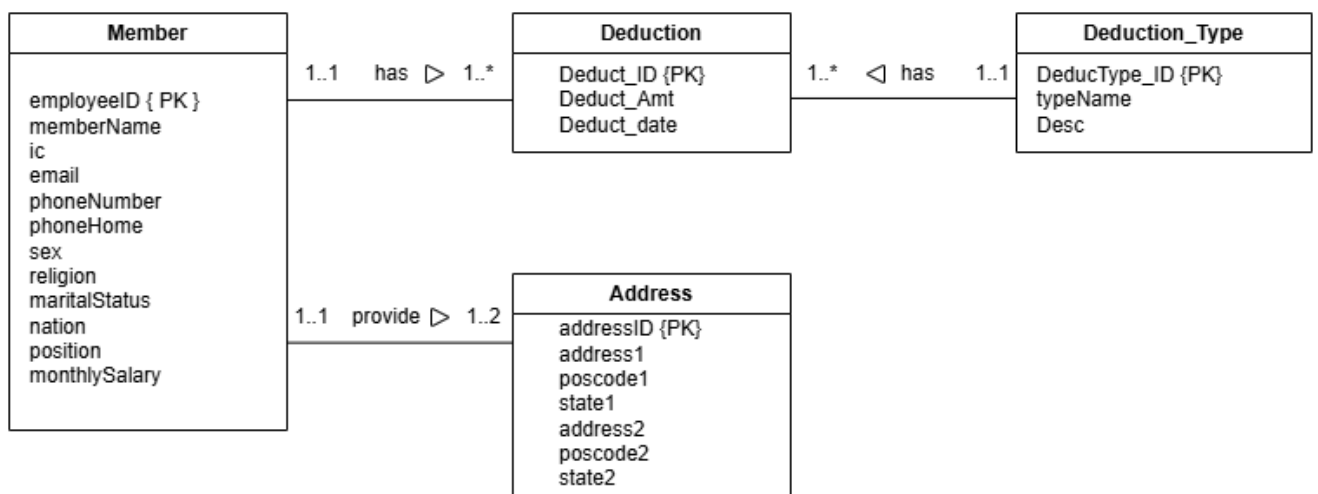
The salary update module is designed to enhance the functionality of the payroll system by allowing the admin to dynamically change, create and manage the deduction type. Admin will be able to choose the deduction name, description, amount and whether the deduction applies to a specific member. This module interacts easily with the existing deduction history log, ensuring that all changes are documented with information such as the date and the time of the adjustments.

Section B (Database Planning And Design)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

Entity Name	Attributes	Description	Data Type & Length	Nulls	Multi-Valued
Member	employeeID (PK)	Unique ID for the member	Int (4)	No	No
	memberName	Name of member	Varchar (50)	No	No
	ic	Number of identity card of member	Int (12)	No	No
	email	Email address of member	Varchar (255)	No	No
	phoneNumber	Mobile phone number of member	Varchar (11)	No	No
	phoneHome	Home phone number of member	Varchar (10)	No	No
	sex	Gender of the member	Varchar (10)	No	No
	religion	Religion of the member	Varchar (30)	No	No
	maritalStatus	Marital Status of member	Varchar (20)	No	No
	race	Race of the member	Varchar (30)	No	No
	position	Job position of member	Varchar (40)	No	No

	monthlySalary	Monthly salary of member	Int (10)	No	No
Address	addressID (PK)	Unique ID for address	Int (10)	No	No
	address1	Address for home address	Varchar (255)	No	No
	poscode1	Postcode for home address	Int (5)	No	No
	state1	State for home address	Varchar (50)	No	No
	address2	Address for office address	Varchar (255)	No	No
	poscode2	Postcode for office address	Int (5)	No	No
	state2	State for office address	Varchar (50)	No	No
Deduction_Type	DeducType_ID (PK)	Unique ID for deduction type	Int (10)	No	No
	typeName	Name of deduction type	Varchar (50)	No	No
	Desc	Description of deduction type	Varchar (255)	No	No
Deductin	Deduct_ID (PK)	Unique ID for deduction	Int (10)	No	No

	Deduct_Amt	Amount of deduction	Int (10)	No	No
	Deduct_date	Date of deduction	Date	No	No

2.4 Relational Database Schema

Member (employeeID, memberName, ic, email, phoneNumber, phoneHome, sex, religion, DOB, maritalStatus, nation, address1, address2, position, monthlySalary)

PK: employeeID

Address (addressID, employeeID, address1, poscode1, state1, address2, poscode2, state2)

PK: addressID

FK: employeeID references Member (employeeID)

Deduction_Type (DedecType_ID, typeName, Desc, Amoun_Type, Default_Value)

PK: DedecType_ID

Deduction (Deduct_ID, Deduct_Amt, Deduct_date, DedecType_ID, employeeID)

PK: Deduct_ID

FK: DedecType_ID references Deduction_Type (DedecType_ID)

FK: employeeID references Member (employeeID)

2.5 Normalization

1NF

Member (employeeID, memberName, ic, email, phoneNumber, phoneHome, sex, religion, DOB, maritalStatus, nation, address1, address2, position, monthlySalary)

PK: employeeID

Address (addressID, employeeID, address1, poscode1, state1, address2, poscode2, state2)

PK: addressID

FK: employeeID references Member (employeeID)

Deduction_Type (DedecType_ID, typeName, Desc, Amoun_Type, Default_Value)

PK: DedecType_ID

Deduction (Deduct_ID, Deduct_Amt, Deduct_date, DedecType_ID, employeeID)

PK: Deduct_ID

FK: DedecType_ID references Deduction_Type (DedecType_ID)

FK: employeeID references Member (employeeID)

2NF

- No partial dependency

Member (employeeID, memberName, ic, email, phoneNumber, phoneHome, sex, religion, DOB, maritalStatus, nation, address1, address2, position, monthlySalary)

PK: employeeID
CHERYL CHEONG KAH VOON_SECP2523_AA_PHASE1

Address (<u>addressID</u>, employeeID, address1, poscode1, state1, address2, poscode2, state2) PK: addressID FK: employeeID references Member (employeeID)
Deduction_Type (<u>DedecType_ID</u>, typeName, Desc, Amoun_Type, Default_Value) PK: DedecType_ID
Deduction (<u>Deduct_ID</u>, Deduct_Amt, Deduct_date, DedecType_ID, employeeID) PK: Deduct_ID FK: DedecType_ID references Deduction_Type (DedecType_ID) FK: employeeID references Member (employeeID)

3NF

- No transitive dependency

Member (<u>employeeID</u>, memberName, ic, email, phoneNumber, phoneHome, sex, religion, DOB, maritalStatus, nation, address1, address2, position, monthlySalary) PK: employeeID
Address (<u>addressID</u>, employeeID, address1, poscode1, state1, address2, poscode2, state2) PK: addressID FK: employeeID references Member (employeeID)

Deduction_Type (<u>DedecType_ID</u>, typeName, Desc, Amoun_Type, Default_Value) PK: DedecType_ID
Deduction (<u>Deduct_ID</u>, Deduct_Amt, Deduct_date, DedecType_ID, employeeID) PK: Deduct_ID FK: DedecType_ID references Deduction_Type (DedecType_ID) FK: employeeID references Member (employeeID)